(2) A

Net profit per car = \$ 70,000 × 25% = \$ 17,500

Q = # of units sold each year from year 6-20

YR 0	YR 1 – 5	YR 6-20
2013	2014-2018	

LINEAR GROWTH

$$\$20\ B = \underbrace{\sum_{t=1}^{5} \frac{\pi_t}{(1+r)^t}}_{(A)} + \underbrace{\sum_{t=6}^{20} \frac{\pi_t}{(1+r)^t}}_{(B)} \quad \text{— (1)}$$

YEAR	1	2	3	4	5
d	0.952	0.907	0.864	0.823	0.784

ANNUITY FACTOR YR 6-20 (FROM PRECALC) $= \frac{1-(1+r)^{-n}}{r} = \frac{1-(1.05)^{-15}}{0.05} = 10.38$

DISCOUNT ANNUITY FACTOR TO PRESENT $= \frac{10.38}{(1.05)^5} = \underline{8.13}$

(A) $= PV_{1-5} = 17500\Big[20{,}000(0.952) + 20000 + 25\%(Q-20000)(0.907)$

$+ 20000 + 50\%(Q-20000)(0.864)$

$+ 20000 + 75\%(Q-20000)(0.823)$

$+ Q(0.784)$

(B) $= PV_{6-20} = 17500 * Q * \underline{8.13}$

FROM (1) AND (A) & (B)

$\$20\ B = 794{,}586{,}618 + 142{,}275\,Q$

$$Q = \frac{[\$20 - 0.794586618] \times 10^9}{142{,}275}$$

$$\boxed{Q = 109{,}926 \text{ units.}}$$